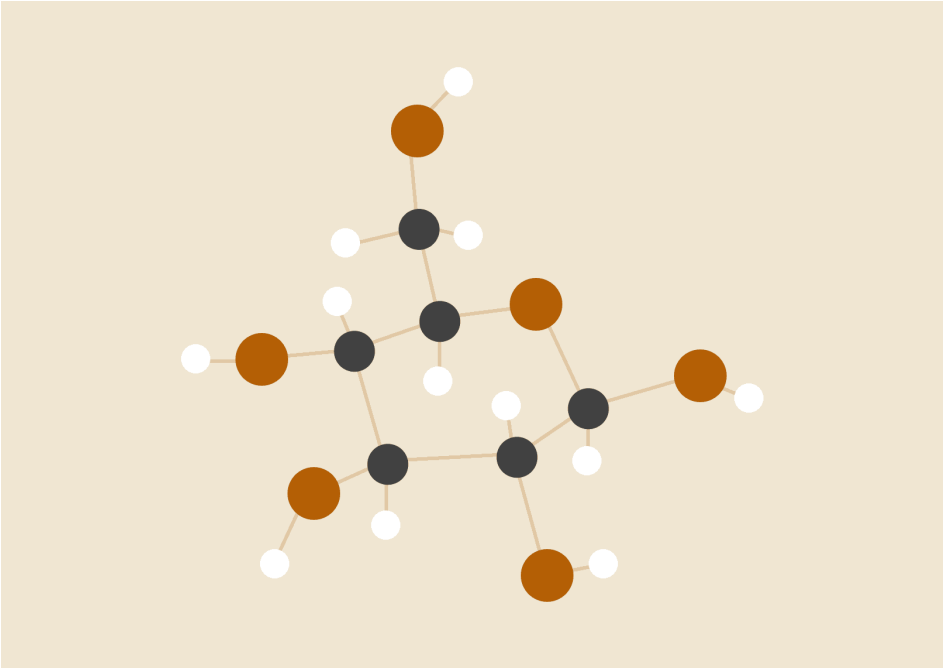
CropSmart NAFSI Track 1: Predictive Modeling for Agricultural Resilience Team: GoodFellas

Abhipsa
Amey Sameer Hengle
Sidharth Mathur



1. Pipeline Overview and Dataset Description

We focused our entire analysis on Iowa (FIPS code 19), which sits at the heart of the U.S. Corn Belt and provides the highest density of corn-soybean rotation data in the country. All three CropSmart datasets were accessed programmatically through the WPS GetFileByFips endpoint, which returns county or state-clipped GeoTIFF files at their native resolution. We did not download data manually, every fetch is reproducible from the notebook by running the data access cells.

The pipeline flows in four sequential stages. First, we download and align the data: CDL rasters at 30m, NDVI weekly composites at 250m, and SMAP anomaly layers at 9km are all reprojected to a common coordinate reference system (EPSG:5070, Albers Equal Area) using rasterio's reproject function with nearest-neighbor resampling for categorical data and bilinear for continuous fields. Second, we run Task 1 and Task 2 together since both depend on the CDL stack. Third, Task 3 uses the SMAP anomaly product with a CDL-derived cropland mask. Finally, Task 4 builds on outputs from all three prior tasks to construct a feature matrix for the Random Forest classifier.

One important note on the SMAP data: CropSmart's WPS service exposes a pre-computed weekly anomaly layer (SMAP-9KM-ANOMALY-WEEKLY-SUB) rather than raw soil moisture. This anomaly is already computed relative to a long-term climatological baseline maintained within the CropSmart platform. We used this directly rather than re-computing our own baseline from raw SMAP data, which means our anomaly maps are internally consistent with the CropSmart system's own reference climatology. For the 2019 event analysis, this is appropriate because the platform's baseline spans multiple years of SMAP data.

2. Task 1: NDVI Time Series Analysis

We extracted weekly NDVI profiles for corn and soybean across Iowa for the 2021 growing season (weeks 14 through 44, approximately April through November). To avoid contamination from mixed pixels at field edges, we applied a purity filter: a 250m NDVI pixel was only included in the corn mask if more than 60% of the underlying 30m CDL pixels were classified as corn (and equivalently for soy). After filtering, we retained roughly 719,000 pure corn pixels and 547,000 pure soy pixels, which is a large enough

sample that the statistics are stable.

Both crops follow the classic warm-season phenological pattern. Starting from early April, NDVI values for both crops sit around 0.65-0.70, reflecting pre-existing winter cover and bare soil after tillage. There is a notable dip in mid-May (around weeks 18-20) in both curves. This is real and expected: it corresponds to the period after planting when the canopy has not yet closed and soil background becomes visible to the MODIS sensor. Corn planters typically hit Iowa fields in late April through mid-May, and soybean planting follows a couple of weeks behind : this timing offset is visible in the slightly later dip for soy.

From June onward both crops green up rapidly. Peak NDVI for corn reaches approximately 0.95 around late July (week 30), while soybean peaks slightly higher at 0.96 and about one to two weeks later, around early August. The later peak for soy reflects its longer vegetative growth period before pod fill. The IQR bands are noticeably tighter for corn than for soy during the peak greenness period, suggesting more uniform management practices in Iowa corn production relative to soybean.

The harvest signal is clear in both crops: NDVI drops sharply after week 38 (late September for corn, mid-October for soy). Corn senescence is faster and steeper, reflecting the mechanical nature of corn drydown versus soybean's more gradual maturation. By late October both crops are back near 0.5-0.6, reflecting post-harvest stubble and bare soil.

The key phenological features that best distinguish the two crops are: (1) the timing of the mid-season dip, which occurs roughly 1.5 weeks earlier for corn; (2) the timing of peak NDVI, which is earlier for corn; and (3) the slope of the post-peak decline, which is steeper for corn. These three features turn out to be directly reflected in the Task 4 model's most important variables (ndvi_peak_week, ndvi_integral).

3. Task 2: Crop Rotation Pattern Identification

We analyzed 10 years of CDL data (2013-2022) for Iowa to classify every corn/soy pixel into one of four rotation categories. Our quantitative criterion for regular rotation was strict: a pixel must have at least 7 out of 9 possible corn-to-soy or soy-to-corn transitions, with at least 3 years of corn and 3 years of soy in the record. This threshold ensures we

are capturing true alternating rotation rather than a field that happened to switch once.

Results show that Iowa is overwhelmingly a rotation state. About 60.9% of corn/soy cropland (roughly 6.22 million hectares) falls into the regular corn-soy rotation class. Continuous corn accounts for 10.4% (1.06 million ha), and continuous soybean is almost nonexistent at 0.2%. The remaining 28.5% (2.91 million ha) is classified as irregular, meaning the field grows corn or soy but not in a consistent alternating pattern.

Looking at the spatial map, regular rotation pixels (shown in blue) are distributed fairly uniformly across most of Iowa, which aligns with the state's reputation as the epicenter of corn-soy rotation. Continuous corn clusters appear more frequently in the northern and northwest portions of the state : this is consistent with known patterns where higher productivity soils and access to tile drainage support continuous corn without the yield drag that typically makes rotation economically necessary. The irregular class tends to cluster near the edges of major rivers and in the southern tier of counties, where flooding risk, soil variability, and smaller farm operations introduce more year-to-year flexibility.

The proportion of cropland under regular rotation is high enough that it would be a reliable prior for any prediction model operating in Iowa : essentially, if you don't know what a field grew last year, betting on the alternate crop from two years ago is a reasonable starting point.

4. Task 3: Soil Moisture Anomaly Analysis

We analyzed SMAP weekly soil moisture anomalies for Iowa across the 2019 growing season (weeks 12-32, mid-March through early August). The 2019 Midwest flooding event was one of the most significant agricultural weather disruptions on record in this region, caused by a combination of above-normal snowmelt, saturated soils from fall 2018, and a series of heavy rainfall events beginning in March.

The anomaly time series over Iowa cropland shows persistently positive values throughout the entire period analyzed. From week 12 (mid-March) through week 26 (late June), the cropland-averaged anomaly ranges from +0.08 to +0.17 m3/m3 above the baseline. This is a very large positive anomaly : for context, the SMAP L4 product's typical soil moisture range across Iowa is roughly 0.20-0.45 m3/m3, so anomalies of +0.15 represent a substantial fraction of the usable water range.

The spatial maps confirm that the wet anomaly was most severe in the western two-thirds of Iowa, particularly along the Missouri River corridor and in the Des Moines Lobe region : both areas that experienced record flooding. Eastern Iowa shows relatively smaller anomalies.

The agricultural implications were severe. Corn planting in Iowa is ideally completed by early May. In 2019, saturated soils kept equipment off fields well into June in many counties, forcing late planting that pushed crop development into a shortened growing season. Fields that were planted late showed reduced yield potential because corn cannot accumulate sufficient growing degree days before the first frost if planting is delayed past mid-June. The anomaly time series still shows values around +0.10 m3/m3 in early July (week 27), which overlaps with early corn tasseling : excessively wet conditions at this stage can impede pollination by encouraging nitrogen leaching and root disease pressure.

By late July and early August (weeks 30-32) anomalies begin declining toward zero, suggesting soils were finally drying enough for normal late-season crop development. Our SMAP analysis confirms that the 2019 event was not a brief pulse but a sustained season-long hydrological anomaly that affected essentially all stages of the corn growing calendar in Iowa.

5. Task 4: Crop Mapping Prediction ModeRESULTS

Our feature matrix draws from three sources: CDL historical sequences, NDVI phenological summaries, and rotation pattern derived aggregates. For each 30m pixel sampled at an 8x stride (equivalent to roughly 240m spatial sampling), we extracted 44 features including per-year binary crop indicators for 2013-2022, cumulative corn/soy/wheat counts over the decade, alternating transition frequency, rotation class, the most recent 3-year crop sequence, 3x3 neighborhood composition, and four NDVI phenology metrics (peak NDVI, peak week, green-up week, and growing season integral) derived from 2022 MODIS data.

We trained a Random Forest classifier with 300 trees and max depth 20, using balanced class weights to address the extreme scarcity of winter wheat pixels in Iowa. Training

used CDL 2013-2022 features with CDL 2023 as the held-out test label. After filtering to pixels with at least 2 years of corn or soy in the most recent 5 years, the valid dataset included roughly 1.67 million pixels, split 80/20 for train/test.

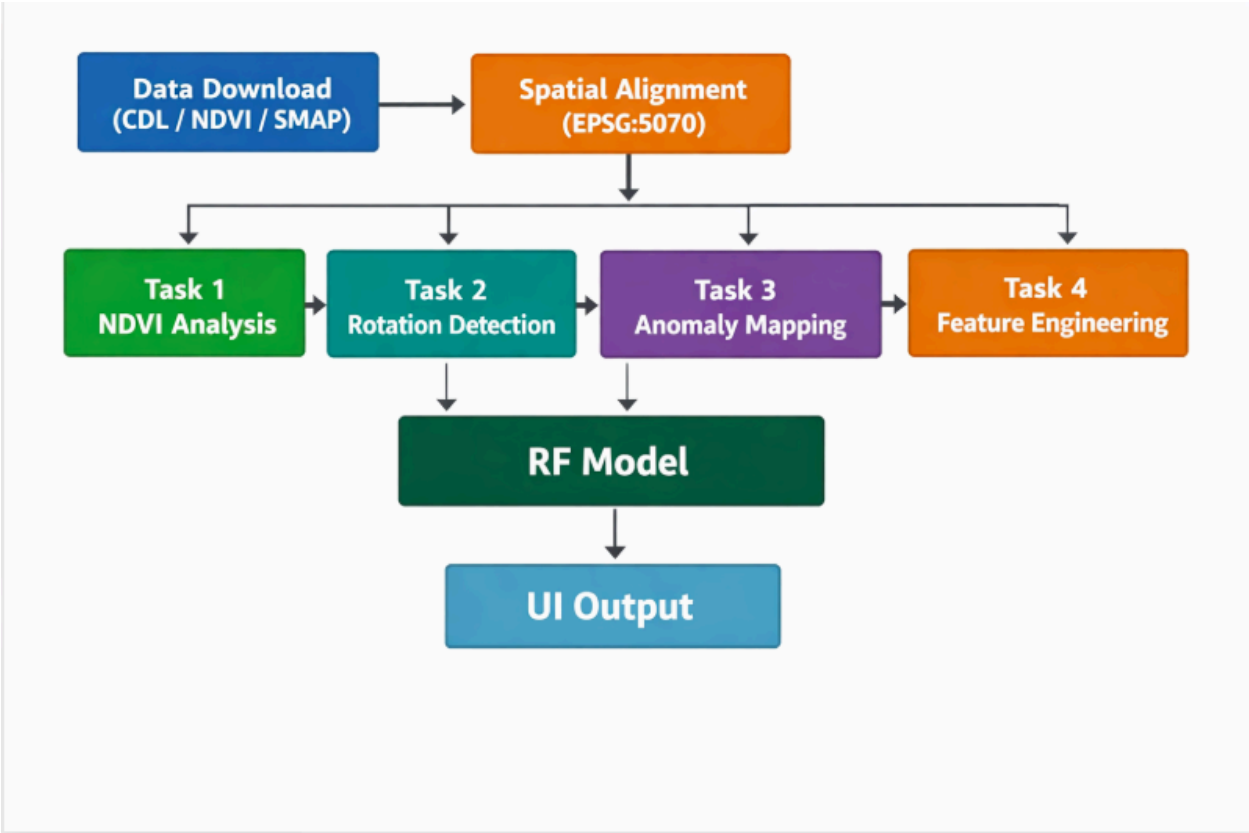
Results: Overall accuracy of 86.7%, with corn F1 of 0.89, soybean F1 of 0.87, and Other/Crop F1 of 0.69. The winter wheat class performed poorly with F1 of 0.15. This is a known limitation of the Iowa study area: wheat is cultivated on only 877 pixels out of 1.67 million, representing less than 0.05% of the validation set. The balanced class weight forces the model to attempt wheat prediction, but 877 training samples is insufficient to learn a reliable decision boundary. In a state with significant winter wheat coverage (Kansas, Oklahoma), this would not be an issue. The confusion matrix shows that wheat pixels are mostly misclassified as corn or other, which is geographically plausible since small wheat fields in Iowa often share similar neighborhood context with corn fields.

Feature importance analysis reveals a clear hierarchy. The CDL label from 2022 (code_2022) is the single most important feature, which makes intuitive sense: if a field grew soy in 2022, it will most likely grow corn in 2023 under normal rotation. The NDVI integral for 2022 is the second most important, confirming that vegetation greenness patterns carry information beyond the categorical CDL label alone. NDVI-derived features (peak NDVI, peak week, green-up week) collectively account for roughly 25-30% of total feature importance, demonstrating meaningful added value from the vegetation index layer. The soy_2022 and corn_2022 binary features and the alt_transitions count round out the top ten.

Performance across rotation classes is stable. Regular corn-soy pixels achieve OA=0.866 and macro F1=0.646. Irregular pixels actually perform slightly better (OA=0.869, F1=0.661), which suggests the model is not simply memorizing rotation rules : it is using the NDVI and spatial neighborhood features to classify even non-standard fields reasonably well. Continuous corn pixels perform at the same level (OA=0.866, F1=0.647). The small continuous soy class (750 test pixels) achieves the highest macro F1 of 0.805, which partly reflects the limited class diversity in those pixels.

6. Response to Report Questions

Q1: Full pipeline description Covered in Section 1 above.



Q2: NDVI phenological differences between corn and soy

The clearest distinguishing features are the timing of the mid-May green-up dip (approximately 1.5 weeks earlier for corn), the timing of peak NDVI (late July for corn versus early August for soy), and the rate of post-peak senescence (faster and steeper for corn). Across years, the corn greenup-to-peak interval tends to be shorter and more consistent, while soybean shows more interannual variability in peak timing, likely reflecting its greater sensitivity to temperature and day length during pod-setting stages.

Q3: Most informative features for rotation detection

The most informative combination is: (1) the crop code sequence for the most recent 3 years, since rotation is autoregressive : what grew in year N strongly predicts year N+1; (2) the alternating transition count over the full 10-year record, which provides a

noise-robust summary of rotation regularity; and (3) the neighborhood corn and soy composition, since field rotation patterns in Iowa tend to cluster spatially at the township scale due to common landlord practices and drainage district influences.

Q4: Model performance on irregular rotation fields Slightly better than expected. Irregular rotation pixels achieve OA=0.869 and macro F1=0.661, essentially matching the model's performance on regular rotation fields. This is because our feature set includes NDVI phenological signals that provide current-year evidence independent of historical CDL sequences. Even if a field breaks its typical rotation pattern, the NDVI peak week and integral features capture the actual crop growth dynamics for that year and help the model make the right call.

Q5: Combining SMAP anomalies and crop predictions for real-world decisions This is where the analysis becomes most actionable. A county extension agent working in Iowa in spring 2019 could combine our outputs in the following way: first, use the rotation map to generate a prior prediction of what each field should be growing this year based on its history. Then overlay the SMAP anomaly map to identify which of those fields are experiencing saturated conditions severe enough to delay planting or switch intended crop (for example, fields on poorly-drained soils in western Iowa where anomalies exceeded +0.15 m3/m3). The combination allows field-level prioritization: a regular corn-soy rotation field in a high-anomaly zone is a high-priority candidate for a monitoring call, while a field in a low-anomaly zone on well-drained soils can be deprioritized. The model output provides the predicted crop type, the SMAP output provides the stress condition, and together they generate a risk score without requiring the agent to have ground-truth knowledge of every field's planting status.

This is precisely the vision behind the UI we built for this submission. Rather than asking farmers or extension agents to interpret raw geospatial outputs, the interface exposes the model's predictions as decision-ready outputs. The user inputs their field's recent crop history and current observed conditions (soil moisture anomaly, NDVI level, green-up timing), and the model returns: predicted crop type with confidence, a planting timing recommendation, a flood/drought risk level, and a yield outlook. The What-If panel allows the user to ask questions like "what happens to my expected yield outlook if soil moisture stays at this level through July?" or "how does a 3-week planting delay change the predicted crop type probability?" This kind of scenario modeling is something no static map product can provide, and it reflects what CropSmart's digital twin concept is ultimately building toward: a system that translates geospatial science into decisions that

a farmer can act on the same morning they check it.

7. Model Limitations and Generalizability

Three main limitations are worth noting.

1. The model is Iowa-specific. The feature importances reflect Iowa's near-universal corn-soy rotation structure, which would not generalize without retraining to states with more diverse cropping systems (California's specialty crops, the South's cotton-sorghum systems, etc.).
2. The 240m spatial sampling we applied to manage memory during training means the prediction map has coarser spatial resolution than the 30m CDL input. A production deployment would need to either sample more densely or tile the prediction across the raster in chunks.
3. The 2022 NDVI features used in training are from a different year than the 2023 test labels. In a real operational setting, the current-year NDVI would be available in near-real-time, but there is a timing dependency that the model assumes NDVI phenology has been largely observed before prediction is needed (i.e., the model would need to run in August or September of the prediction year, not earlier).